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Network delay inference from additive metrics

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Domain Signal processing, statistics & applied mathematics

Abstract

We use computational phylogenetic techniques to solve a central problem in inferential network monitoring. More precisely, we design a novel algorithm for multicast-based delay inference, that is, the problem of reconstructing delay characteristics of a network from end-to-end delay measurements on network paths. Our inference algorithm is based on additive metric techniques used in phylogenetics. It runs in polynomial time and requires a sample of size only poly(log n). We also show how to recover the topology of the routing tree.

Real-World Impact & Application

Diagnosing Discord & FaceTime Voice Lag: Ever had your voice echo or cut out on a Discord call? This paper treated network travel delays like addition on a tree, enabling network diagnostics to pinpoint which physical internet cable is congested without opening up telecom switches.